Analogue electronics: mixed questions

National 5 · Electronics & control · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 A 12 V supply is connected to a 240 Ω resistor. Calculate the current.
- I = 12 ÷ 240 = 0.05 A

2a) Calculate the total resistance.
- R_T = 50 Ω

2b) Calculate the current.
- I = 10 ÷ 50 = 0.2 A

2c) Calculate the voltage across each resistor.
- 3 V and 7 V

3a) Calculate the total resistance.
- R_T = 75 Ω

3b) Calculate the total current.
- I_T = 12 ÷ 75 = 0.16 A

3c) Calculate the current through the 100 Ω resistor.
- 12 ÷ 100 = 0.12 A

4 A current of 0.5 A flows through a resistor on a 9 V supply. Calculate the resistance.
- R = 9 ÷ 0.5 = 18 Ω

5a) Calculate the total resistance.
- $R_{parallel}$ = 20 Ω; R_T = 40 Ω

5b) Calculate the current.
- I = 18 ÷ 40 = 0.45 A

5c) Calculate the voltage across the 20 Ω resistor.
- V = 0.45 × 20 = 9 V

6a) Calculate the total resistance.
- 6.9 kΩ

6b) Calculate the current.
- I = 12 ÷ 6900 = 1.74 mA

6c) Calculate the voltage across R_2.
- V = 0.00174 × 4700 = 8.2 V

7a) Calculate the total resistance.
- $1/R_T$ = 0.1 + 0.05 + 0.025 = 0.175; R_T = 5.71 Ω

7b) Calculate the total current.
- I_T = 12 ÷ 5.71 = 2.1 A

7c) Calculate the current through the 20 Ω resistor.
- 12 ÷ 20 = 0.6 A

8 A current of 2 A flows through a 15 Ω resistor. Calculate the supply voltage.
- V = 2 × 15 = 30 V

9a) Calculate the total resistance.
- 500 Ω

9b) Calculate the current.
- 20 ÷ 500 = 0.04 A

9c) Calculate the voltage across the 250 Ω resistor.
- 0.04 × 250 = 10 V

10a) Calculate the total resistance.
- (8 × 12) ÷ 20 = 4.8 Ω

10b) Calculate the total current.
- 6 ÷ 4.8 = 1.25 A

11a) Calculate the total resistance.
- $R_{parallel}$ = 50 Ω; R_T = 100 Ω

11b) Calculate the current.
- 10 ÷ 100 = 0.1 A

11c) Calculate the current through one of the 100 Ω resistors.
- V_{50} = 5 V, so 5 V across the pair
- I = 5 ÷ 100 = 0.05 A

12a) Draw a circuit to give 25 Ω.
- The two 50 Ω resistors in parallel

12b) Prove your answer with a calculation.
- (50 × 50) ÷ 100 = 25 Ω

13 A 24 V supply is connected to a 3 kΩ resistor. Calculate the current in mA.
- 24 ÷ 3000 = 0.008 A = 8 mA

14a) Calculate the total resistance.
- (1000 × 2000) ÷ 3000 = 667 Ω

14b) Calculate the total current.
- 9 ÷ 667 = 0.0135 A

15a) Calculate the total resistance.
- $R_{parallel}$ = 20 Ω; R_T = 30 Ω

15b) Calculate the current.
- 12 ÷ 30 = 0.4 A

15c) Calculate the voltage across the 10 Ω resistor.
- 0.4 × 10 = 4 V

15d) Calculate the current through the 30 Ω resistor.
- $V_{parallel}$ = 12 − 4 = 8 V
- I = 8 ÷ 30 = 0.27 A

16a) Calculate the total resistance.
- 1970 Ω

16b) Calculate the current.
- 9 ÷ 1970 = 4.57 mA

17a) Calculate the total resistance.
- (20 × 30) ÷ 50 = 12 Ω

17b) Calculate the total current.
- 12 ÷ 12 = 1 A

17c) Calculate the current through the 20 Ω heater.
- 12 ÷ 20 = 0.6 A

18 A circuit has a total resistance of 40 Ω: a 10 Ω resistor in series with a parallel section. Calculate the resistance of the parallel section.
- 40 − 10 = 30 Ω

19a) Calculate the total resistance.
- 1250 Ω

19b) Calculate the total current.
- 10 ÷ 1250 = 8 mA

19c) Calculate the current through the 2 kΩ resistor.
- 10 ÷ 2000 = 5 mA

20a) Calculate the total resistance.
- $R_{parallel}$ = 15 Ω; R_T = 30 Ω

20b) Calculate the current.
- 6 ÷ 30 = 0.2 A

20c) Calculate the voltage across the parallel section.
- 0.2 × 15 = 3 V

20d) Calculate the current through one 30 Ω resistor.
- 3 ÷ 30 = 0.1 A